

# Weak Bonds? Competing Regional Customs Unions and the German Market before the Zollverein

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## Abstract

In 1828, three competing regional customs unions formed simultaneously in the German Confederation, later merging into the Zollverein in 1834. The Zollverein is well studied; its three immediate precursors are not. Existing work treats 1815–1833 as a homogeneous pre-Zollverein baseline and does not separately test the Prussian–Hessian (PHC), South German (SGCU), and Central German (CGCU) unions’ own market-integration performance. This paper measures each separately. Using a dyadic panel of annual grain prices for cities across 17 German states over 1818–1833, I combine within- and cross-union price-gap regressions, spatial autocorrelation statistics, and event-study timing diagnostics, with dyad-robust inference. No union shows internal convergence: the Central German Commercial Union’s within-union coefficient is indistinguishable from zero, the South German union’s is positive but rests on four city pairs, and the Prussian–Hessian system cannot be estimated. Border-effect point estimates are negative for all three. Throughout, geographic proximity predicts which prices co-move better than shared membership does. Particular border combinations show mixed results: pairs along the CGCU–PHC frontier show gaps wider by 2.8 per cent, the only robust cross-union estimate. Read against the founding treaties, the pattern points to pre-existing institutional baselines rather than to the treaties: the Prussian–Hessian system extended Prussia’s own 1818 customs law to new partners, while the Central German members committed to freezing, not eliminating, existing transit tolls, and to not acceding to either rival. This is the first price-based measurement of each union’s individual performance, and the first price evidence on the competitive phase of the Zollverein’s formation.

**Keywords:** Zollverein; market integration; customs union; border effects; price convergence; spatial autocorrelation; economic history

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# 1 Introduction

Regional trade agreements can serve as building blocks of wider integration or as stumbling blocks against it (Baldwin, 1993; Bhagwati, 1993; Viner, 1950). The debate is usually conducted on modern data, where trade agreements are numerous, overlapping, and difficult to isolate from one another. The German Confederation of the late 1820s is a cleaner case: the first episode in which several customs arrangements formed in explicit competition with one another, on the same territory, in the same year. In 1828, three blocs emerged among the German states: the Prussian–Hessian system (PHC), a South German customs union of Bavaria and Württemberg (SGCU), and a Central German Commercial Union (CGCU) assembled explicitly to resist absorption into either. These consolidated into the Zollverein in 1834.

The Zollverein itself is intensively studied; its three competing precursors are not. Quantitative work on German market integration treats 1815–1833 as a homogeneous pre-Zollverein baseline (Keller and Shiue, 2014; Shiue, 2005), while the historiography of the 1828 system is rich but qualitative (Hahn, 1984; Henderson, 1934; Price, 1949). The gap is widest on the theoretical side: Ploeckl (2015, 2021) provides a coalition-externality account of the formation process in which the CGCU figures as a defensive blocking coalition, its treaty content “focused on preserving the trade route access of member states and hindering the enlargement of the two newly created customs unions”, but, as he notes, no price evidence exists for what that competition cost. This paper supplies the price evidence: the first separate, price-based examination of each 1828 union’s market-integration performance, and in particular of the competitive frontier between the CGCU and the Prussian system, where the blocking commitment applied most directly. The approach is descriptive: dyadic grain price-gap regressions, spatial autocorrelation statistics, and event-study timing diagnostics on an annual panel of German city grain prices, 1815–1833.

The analysis yields three findings. First, geography governs integration throughout the period: spatial weighting schemes built on distance, nearest neighbours, or contiguity outperform one built on shared union membership in every year, and price gaps decay with distance without breaking at union boundaries. Second, no union integrated its own interior: not one within-union coefficient in this paper is negative. Third, the only boundary with a robust price-gap difference lies between two unions rather than inside any of them: pairs straddling the frontier between the Central German coalition and the Prussian system show price gaps wider by roughly 2.8 per cent, the sole cross-union estimate that survives the full distance sweep.

The third finding is crucial. In the coalition-externality account of the Zollverein’s formation, what drives blocs to merge is not the gain from membership but the cost each bloc imposes on the others: blocking accession raises the price of remaining outside, and

it is that pressure which eventually brings members in (Ploeckl, 2015, 2021). I find no evidence that the 1828 arrangements integrated markets, but the results are consistent with costs being imposed and with the coalition externality being real and detectable.

The paper therefore contributes three things: separate price-based measurement of each 1828 arrangement, rather than absorption into a homogeneous pre-Zollverein baseline; the first application of spatial price statistics to this historiography at the level of individual cities; and a quantification, at one frontier, of the competitive pressure that the formation literature treats as the seed of the Zollverein.

The remainder of the paper proceeds as follows. Section 2 describes the historical setting: the fragmentation of the German customs landscape after 1815, the Prussian customs law of 1818, and the three arrangements of 1828, including a close reading of the CGCU treaty's commercial provisions. Section 3 relates the paper to the quantitative integration literature and the formation historiography. Sections 4 and 5 describe the price panel and the empirical approach. Section 6 establishes the geographic backdrop; Section 7 presents the regression results and their timing diagnostics. Section 8 interprets the pattern through the unions' treaty content, and Section 9 concludes.

## 2 Historical Background

### 2.1 Fragmentation after 1815 and the 1818 Prussian customs law

The German Confederation of 1815 comprised 39 sovereign states, each with its own customs regime, and many with internal tariff lines besides. Rivers carried much of the bulk trade, and their tolls were governed separately from state customs policy by commissions established at the Congress of Vienna (Ploeckl, 2021).

Prussia moved first and unilaterally. The customs law of 26 May 1818 abolished all internal Prussian tariff lines and erected a single external border system with moderate nominal rates (Hahn, 1984; Henderson, 1939). For the small and middle states lying between the two Prussian territorial blocks, the 1818 system raised the cost of trading across Prussian territory. Hesse-Darmstadt, Hesse-Kassel, and Nassau each attempted independent tariff reforms in response (Nassau in 1822, the two Hesses in 1824) at rates below Prussia's own. Nassau's proved fiscally workable; the Hessian attempts largely failed (Hahn, 1982). These economies were heavily exposed to trade across the new Prussian line: Rheinhessen exported roughly a third of its grain production, and Hesse-Kassel was not self-sufficient in grain. The 1828 agreements thus opened after a decade of failed unilateral adjustment and negotiation in response to a Prussian tariff schedule.

## 2.2 The three competing unions of 1828

Within a single year, three arrangements partitioned the German customs landscape (Figure 1). They differed sharply in institutional content, and those differences bear on how their outcomes should be interpreted.

The **Prussian–Hessian Customs Union (PHC)** began with Hesse-Darmstadt’s accession treaty of 14 February 1828. It was less a negotiated union than an extension of Prussia’s 1818 customs system to a new partner: Prussia’s external tariff, Prussia’s administration, with revenue shared by population (Henderson, 1934; Ploeckl, 2015). Contemporaries and later historians alike noted that the agreement had no obvious commercial upside for Prussia; it created no tariff-free connection between the two main Prussian territorial blocks and controlled no new trade route. It was extremely generous towards Hesse-Darmstadt in financial terms and its value for Prussia itself was strategic (Ploeckl, 2015, p. 288).

The **South German Customs Union (SGCU)** of Bavaria and Württemberg was a genuine customs union among equals: a common external tariff, dissolved internal customs posts, and joint institutions (Helmreich, 1949; Henderson, 1934). Both of these states had already reformed their customs systems during Napoleonic times from internal dues to an external system, and negotiated further with other middle and small German states in response to the establishment of the Prussian Customs Law as a way to prevent Prussian influence from expanding (Price, 1949). In the end, only these two states formed the union. Part of Bavaria, the Palatinate, would initially be excluded (de Martens, ed, 1830, p. 530) and be included only at a later point (Kunz, 2008).

The **Central German Commercial Union (CGCU)** (Mitteldeutscher Handelsverein) was concluded at Cassel on 24 September 1828 by a heterogeneous group spanning Saxony, Hanover, Hesse-Kassel, the Thuringian states, Bremen, and Frankfurt, covering the fair cities and the roads between the south and the North Sea ports. Its institutional content was different in kind: it was “neither a customs union nor did it create a free trade area between its member states” (Ploeckl, 2015, p. 289).<sup>1</sup> Its two operative commitments were (i) to preserve members’ route access by undertaking not to raise existing transit tolls and announcing a joint road programme, and (ii) a legal commitment by every member *not to join either other union* for the treaty’s duration. Hahn’s summary of the Cassel convention is that “the negative objective predominated,” and that the treaty “created no unified tariff system in contrast to the two existing customs unions and brought about no complete abolition of internal trade barriers” (Hahn, 1984, pp. 50–51). The distinction matters for what follows: the coalition undertook not to *raise* existing tolls, not to remove

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<sup>1</sup>It does not fulfill the criteria for a customs union set out by Viner (1950). However, it is often referred to as one in the historical literature, and Viner’s own appendix of customs union contracts and treaties includes accessions to the CGCU. I therefore refer to it throughout as a “union” (CGCU) for expositional symmetry with PHC and SGCU.

them. The road programme never materially began (Ploeckl, 2015); the no-join clause was binding and was observed. Prussian policy took the coalition seriously in exactly these terms: Finance Minister Motz’s conditions for the subsequent negotiations with the south listed “the elimination of all its negative effects on Prussia” as a named objective (Ploeckl, 2015, p. 285), and from autumn 1828 Prussia financed bypass roads through the Thuringian states, which were toll-free and in exchange for undertakings to join the Prussian union after 1834, in order to route transit away from Hesse-Kassel (Hahn, 1984).

The geography of the coalition gives this rivalry its economic content. The CGCU territory lay as a belt between the two Prussian territorial blocks, and Hesse-Kassel in particular levied transit tolls on goods moving between Prussia’s Rhenish provinces and its eastern lands, which was a wedge on Prussian internal commerce that supplied an appreciable share of Kurhessian state revenue (Hahn, 1984). The competitive frontier was therefore not only a line across which accession was contractually blocked; it was also the margin on which one side collected rents from the other’s internal trade, and on which both sides were actively re-routing traffic. Austrian backing for the coalition should be qualified in the same spirit: Metternich supported it declaratorily, but the material commercial concessions its members hoped for from Austria, Britain, and France did not arrive (Hahn, 1984).

The system was short-lived. Hesse-Kassel, which was wedged between the two Prussian blocks, and whose southern population blamed tariff policy for economic misery by 1830, defected to the Prussian system by treaty of 25 August 1831, physically reconnecting the two Prussian territories and breaking the CGCU’s geographic position (Hahn, 1982; Ploeckl, 2015). This shift split the former CGCU into two parts, and completely blocked access to the north and thereby the coastline to all southern states. This led to negotiations and capitulation by other states, with the March 1833 treaties between the Prussian system and the SGCU creating the Zollverein, in force from 1 January 1834, where this paper’s window closes.

## **2.3 The CGCU treaty’s commercial provisions**

Because the CGCU’s commercial content bears directly on performance, we examine it in the treaty’s own text, and I examine the provisions one by one (Great Britain. Foreign and Commonwealth Office., 1829, p. 1266-1274). Article XIV nominally created internal free trade in provisions and fuel from 1 January 1829: a twenty-six item duty-free list headed by wheat, rye, spelt, barley, and oats, running on through pulses, potatoes, fodder, and household fuel. Four qualifications limited it.

First, the exemption applied only to goods carried from one member state to another “without touching a Foreign State”; a contiguity condition the coalition’s geography could not generally satisfy, since many commercially significant intra-union routes crossed

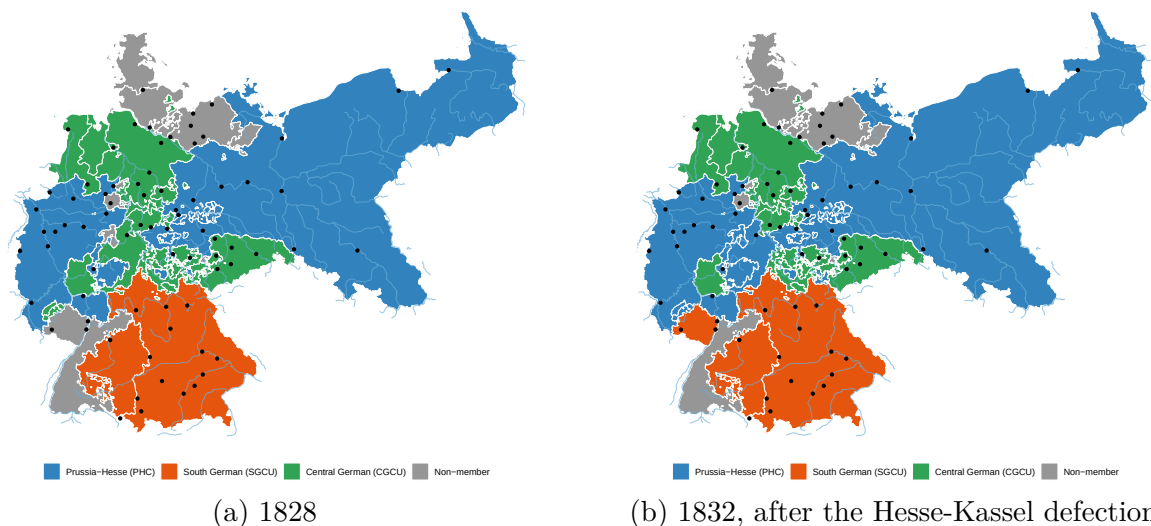


Figure 1: The three customs blocs of 1828. Panel (a) shows the configuration established in 1828: the Prussian–Hessian system (PHC), the South German Customs Union (SGCU), and the Central German Commercial Union (CGCU). Panel (b) shows 1832, after Hesse-Kassel’s accession to the Prussian system reconnected Prussia’s two territorial blocks. Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Prussian or PHC territory.

Second, for the corn and pulse items (numbers 1 to 11, which include every grain in our sample), the concession ran through a single channel. Article XIV.3 (Great Britain. Foreign and Commonwealth Office., 1829, p. 1272) states that the exemption is “merely to facilitate as much as possible the intercourse and traffick” in these goods, and expressly not to serve mercantile speculation; it confines duty-free entry to corn offered for sale at weekly markets under certificates of origin, with retailers limited to certain quantities exempt from the certificate requirement. Grain moving outside that channel paid the ordinary import and transit duties.

Third, transit was capped rather than abolished. Article XIV.2 frees the listed articles from transit duty, but provides that on items 1 to 11, “where a transit duty has already existed, a duty of half a good-groschen per cwt. may be levied.” For grain, then, the treaty set a ceiling on pre-existing transit charges; it did not remove them.

Fourth, internal taxation was untouched: duties on internal commerce and consumption payable on these articles under each member’s own regulations are “not annulled or repealed” by the article. Article XIII further allowed members to prohibit corn exports on political grounds.

The grain provision is therefore commercial language whose geographic precondition the coalition’s own membership could not generally meet, and whose operative concession was confined to weekly-market sales at a capped transit rate. What the treaty did bind was external: the commitment not to join either rival union. In the terms that matter

for prices, the CGCU neither removed barriers among its members nor erected new ones toward Prussia. The Prussian tariff of 1818 was already in place; what the coalition added was a ceiling on its members' existing transit charges, and a commitment to remain outside that tariff.

### 3 Related Literature

**Market Integration and Methods.** The dyadic price-gap design used here is standard in the market integration literature (Chilosi et al., 2013; Federico, 2024; Jacks, 2005), as are its known limitations: price gaps bound trade costs only under assumptions about arbitrage that annual data cannot verify (Coleman, 2007), and estimated border effects are sensitive to sample construction and to what distance is allowed to absorb (Coughlin and Novy, 2021). The spatial statistics follow Anselin (1988, 1995). Inference on dyadic data requires allowing arbitrary dependence among all pairs sharing a city (Aronow et al., 2015; Cameron and Miller, 2023). Finally, Wolf and Ritschl (2011) establish that contemporaneous, competing arrangements must be entered separately rather than pooled into a single membership indicator, which is the reason for the three-way decomposition in Section 5.2; Baier and Bergstrand (2004) supply the complementary point that agreements among geographically ill-suited partners need not raise welfare.

**Quantitative German market integration.** A substantial body of work measures German price convergence over the long nineteenth century. Shiue (2005) estimates border effects on absolute wheat price gaps for 1815–1855 and reports a border coefficient of approximately  $-0.046$ , using France, Switzerland, and Austria as comparators. Keller and Shiue (2014) instrument for the non-random sequence in which states acceded and estimate a considerably larger effect than naive specifications yield; a price-gap reduction of about 0.055 log points, precisely because the benefits of membership rose with the union's size. Related work separates the contributions of customs policy, technology, and institutions (Keller and Shiue, 2008, 2020). Albers and Pfister (2023) treat the Napoleonic territorial reorganization as a natural experiment and find that territorial consolidation reduced price gaps by the equivalent of 31 per cent, conditional on the trade reforms that replaced heterogeneous internal duties with unified external tariffs; trade reform, not consolidation alone, is their operative channel. Others locate the decisive integration earlier or elsewhere: Fremdling and Hohorst (1979) date most Prussian internal integration to before 1820, and Uebele (2010, 2011) document north German price co-movement that crosses customs lines rather than following them. Wolf (2009) questions whether the German market was ever integrated in the sense the Zollverein literature assumes. What this literature has in common is its treatment of the years before 1834: 1815–1833 functions as a homogeneous pre-Zollverein baseline against which the Zollverein is measured. The three arrangements

of 1828 are, at most, a sentence. Shiue describes the CGCU as “essentially a defensive agreement among members to commit to not joining either the Prussian/Hesse-Darmstadt or the South German Customs Union” (Shiue, 2005, p. 134), and does not code it. Keller and Shiue (2008) state the limitation directly, noting that liberalizations among German states outside the Zollverein rounds are incorporated only partially and that complete information on grain protection for the period is not available to them.

One anticipated objection should be met here. Keller and Shiue (2014) report that the Zollverein’s trade diversion was “rather limited.” That finding concerns the union’s external tariff against the rest of the world after 1834. The object of this paper is different in both respect and period: barriers among German states during the competitive phase of 1828–1833, before any of the three arrangements had consolidated.

**Formation historiography and coalition theory.** A second literature explains how the Zollverein came about, and it has the mechanism. The narrative histories establish the institutional facts used in Section 2: the CGCU as a defensive coalition without a common tariff or joint administration (Hahn, 1984; Helmreich, 1949; Henderson, 1934, 1939), the failed unilateral reforms of the 1820s (Hahn, 1982), and the German literature’s characterization of the coalition as a counter-union (Linner and Henke, 2014). Dumke (1984) advances the competing fiscal reading, in which revenue rather than commercial strategy drives accession, drawing on the contemporary account of Kühne (1836); Oechelhäuser (1851) is the other main primary source on the tariff schedules. Reckendrees (2023) places the sequence in a framework of stepwise intergovernmental integration, and attributes the Zollverein’s effectiveness to positive institutional content (a common tariff, revenue sharing, and regular conferences) of which the CGCU had none. Ploeckl (2015, 2021) supply the formal account this paper takes as its object: the unions of 1828 as coalitions formed under externalities, with the CGCU’s treaty content directed at preserving members’ route access and blocking the enlargement of its two rivals. He is explicit that the formation sequence itself does not admit the quantitative treatment the outcome literature applies to the Zollverein (Ploeckl, 2015, p. 283), and no price evidence on the competitive phase is offered.

## 4 Data

### 4.1 Sources and processing

The price data are annual grain prices for German cities, 1815–1833, assembled from Albers and Pfister (2024) and supplemented by Jacks (2005) and Shiue and Keller (2007). Different cities feature prices at different frequencies, often using different weight and currency units, as was the case during the period. Frequencies are matched by taking

the averages within a year for cities where the frequency is higher than annual. Where necessary, historical tables and conversions (Gerhard, 2015) are used to convert all units into standardised prices denoted in grams of silver per litre of grain, as is standard in this literature (Albers and Pfister, 2024). In the case of Bremen, which used a gold based currency, the historical gold silver ratio (Officer and Williamson, 2022) is used to convert to silver equivalents.

Historical GIS files are used in order to link cities to the territories and customs unions they were a part of. MPIDR Max Planck Institute for Demographic Research and CGG Chair for Geodesy and Geoinformatics, University of Rostock (2011) provide annual GIS files at the administrative level of the German Confederation from 1815–1870; these are cleaned and coded, and linked with customs treaties of which territories belonged to which unions at various points in time (Kunz, 2008; Viner, 1950). Union membership is coded from the treaties at annual resolution: PHC from 1828, with Hesse-Kassel from 1832 (its accession treaty is of August 1831 but took effect on 1 January 1832, and membership is coded from the effective date), SGCU from 1828, and CGCU from 1828 until dissolution.

The panel covers five grains: wheat, rye, barley, oats, and spelt. The estimation sample applies four restrictions, all imposed before any specification is built. Spelt is dropped for thin coverage, leaving four grains. Years are limited to 1818–1833, to exclude the extreme harvest-failure years at the panel’s start (see Section 4.2); pairs are limited to those within 200 km; and pairs of cities lying in the same state are dropped, since within-state pairs carry no customs boundary of any kind and would otherwise enter the omitted category. What remains is 65 cities forming 429 cross-state pairs, and 18,772 pair–grain–year observations, spanning 17 states and 20 GEO units. The border and frontier regressions additionally exclude pairs that are ever co-members of a union, leaving 60 cities, 334 pairs and 16,263 observations. Part of that reduction is mechanical rather than substantive: pair types are classified from each city’s customs status in 1828, so a city with no price observation in that year cannot be classified, and three Saxon cities (Dresden, Zwickau and Leisnig) fall out of the classified specifications on that ground.

Table 1 reports the composition.

The unit of analysis is the city pair. For each pair, grain, and year I compute the absolute log price gap, the standard dyadic integration measure (Federico, 2024); it is defined formally in equation (1). The 200 km default deserves a comment, since every threshold is varied in robustness (Section 7.3). Overland grain haulage in this period was expensive relative to the value of the cargo, and beyond roughly two hundred kilometres inland cities were generally not in direct arbitrage with one another but linked, if at all, through intermediate markets, so a price gap between two distant cities is not informative about the barrier between them (Coleman, 2007). The distance profile of the gaps in Section 6.2 is consistent with this.

## 4.2 Descriptive overview

Figure 2 summarises the panel. Coverage is stable: the number of cities reporting a price in a given year moves only between 65 and 75 across 1815–1833, so the series that follow are not driven by entry and exit.

Panel (a) plots mean prices, and a common macro trend dominates them. The harvest failure of 1816 drives prices to a peak in 1817 that stands at between 3.2 and 4.5 times the mid-1820s trough, depending on the grain. Prices fall through 1824–1825, which is the trough for every grain, then rise again from 1826 to a secondary peak in 1831 that reaches roughly half the 1817 level, before falling sharply in 1833. The four grains move together throughout: the profile is the same in each panel, and differences are of amplitude rather than timing. The regression window begins in 1818 to exclude the two extreme years, 1816 and 1817; 1818 itself remains elevated, but by then the series is falling and no longer an outlier against the rest of the period. Nothing in panel (a) marks 1828.

Panel (b) plots the coefficient of variation across cities, computed in price levels as the standard deviation over the mean, which makes it comparable across grains. Dispersion is modest and fairly flat, moving mostly between 0.15 and 0.28, with the exception of the initial years. Overall, there seems to be a slight downward trend even when cutting the famine years, which implies that there was slight and gradual integration occurring over this period.

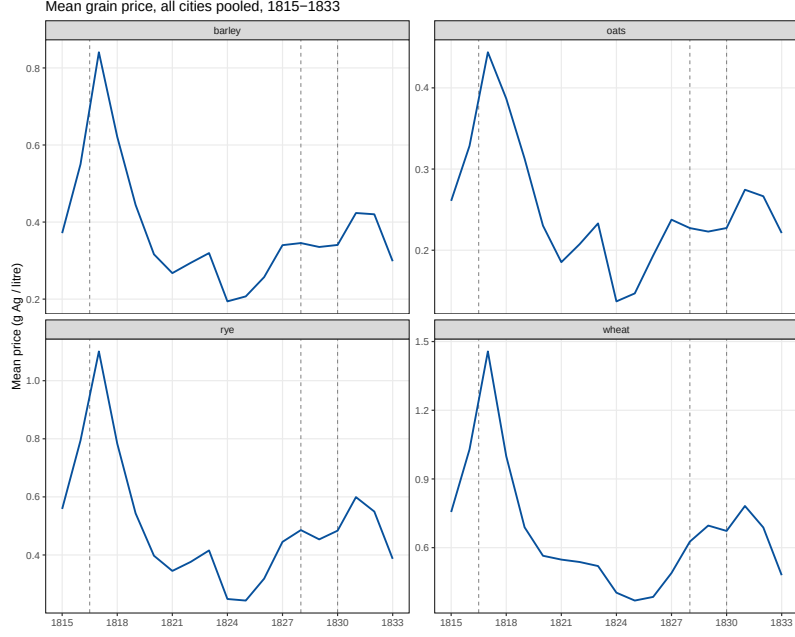
Table 1 gives the sample composition. The pair types are unevenly populated: the CGCU–PHC frontier is the largest category (178 pairs, 39 cities), while the SGCU within-union and SGCU border cells contain four and eight pairs.

## 5 Empirical Approach

The approach is descriptive. I measure whether customs-union membership shows up in bilateral grain price gaps and in the spatial structure of price co-movement. I do not estimate causal effects of membership: the unions were bargained outcomes of the trade patterns the data record (Keller and Shiue, 2014; Wolf and Ritschl, 2011), and no design available in this setting isolates them from that history. The event studies are timing diagnostics on descriptive contrasts: they ask *when* a difference arose, not what caused it.

### 5.1 Price Gaps

Let  $p_{i,g,t}$  denote the price of grain  $g$  in city  $i$  in year  $t$ , measured in grams of silver per litre. For each unordered city pair  $(i, j)$ , grain  $g$ , and year  $t$ , the dependent variable is the



(a) Mean price, all cities pooled



(b) Coefficient of variation across cities

Figure 2: Grain prices, 1815–1833, by grain. Panel (a) plots the mean price across all sample cities in grams of silver per litre. Panel (b) plots the cross-sectional coefficient of variation, computed in price levels as the standard deviation across cities divided by the mean. Dashed vertical lines mark the 1816–17 famine, the union treaties of 1828, and the revolutions of 1830.

absolute log price gap

$$G_{ij,g,t} \equiv \left| \ln p_{i,g,t} - \ln p_{j,g,t} \right| = \left| \ln \left( \frac{p_{i,g,t}}{p_{j,g,t}} \right) \right| \geq 0. \quad (1)$$

Table 1: Sample composition. Pair types in Panel A are classified by customs status from 1828; “mean gap” is the mean absolute log price gap. Estimation sample: 1818–1833, pairs within 200 km, cross-state pairs only, spelt excluded.

|                                       | Pairs | Cities | Obs.   | Mean gap |
|---------------------------------------|-------|--------|--------|----------|
| <i>Panel A: pair types, 1828–1833</i> |       |        |        |          |
| Neither city in a union               | 16    | 12     | 218    | 0.183    |
| Both in CGCU                          | 44    | 17     | 538    | 0.147    |
| Both in SGCU                          | 4     | 5      | 66     | 0.116    |
| Both in PHC                           | 26    | 20     | 250    | 0.140    |
| CGCU border                           | 60    | 21     | 1,210  | 0.167    |
| SGCU border                           | 8     | 8      | 100    | 0.154    |
| PHC border                            | 61    | 34     | 1,266  | 0.179    |
| CGCU–PHC frontier                     | 178   | 39     | 3,110  | 0.151    |
| CGCU–SGCU frontier                    | 16    | 12     | 146    | 0.229    |
| PHC–SGCU frontier                     | 25    | 16     | 280    | 0.193    |
| <i>Panel B: by grain, 1818–1833</i>   |       |        |        |          |
| Wheat                                 | 355   | 58     | 5,156  | 0.179    |
| Rye                                   | 400   | 61     | 5,860  | 0.162    |
| Barley                                | 288   | 53     | 4,157  | 0.183    |
| Oats                                  | 248   | 49     | 3,599  | 0.195    |
| <b>Full sample</b>                    | 429   | 65     | 18,772 | 0.178    |

This is the standard dyadic measure of market integration (Federico, 2024; Jacks, 2005): the proportional distance between two cities’ prices for the same grain in the same year, taken in absolute value because the direction of the gap is not informative about integration. Under a spatial-arbitrage reading,  $G_{ij,g,t}$  is bounded above by the cost of moving grain between the two markets. A lower  $G$  indicates closer integration, so a positive coefficient below means wider gaps.

## 5.2 Specification

The set of possible unions is  $\mathcal{U} = \{\text{PHC}, \text{SGCU}, \text{CGCU}\}$  and  $u_{i,t} \in \mathcal{U} \cup \{\emptyset\}$  is the customs affiliation of city  $i$  in year  $t$ , with  $\emptyset$  denoting no union. Three families of indicators are defined by the pair’s joint customs status in year  $t$ :

$$\text{Within}_{ij,t}^u = \mathbf{1}\{u_{i,t} = u_{j,t} = u\}, \quad \text{both cities in union } u; \quad (2)$$

$$\text{Border}_{ij,t}^u = \mathbf{1}\{u_{i,t} = u, u_{j,t} = \emptyset\} + \mathbf{1}\{u_{i,t} = \emptyset, u_{j,t} = u\}, \quad \text{one city in } u, \text{ one unaffiliated}; \quad (3)$$

$$\text{Frontier}_{ij,t}^{uv} = \mathbf{1}\{\{u_{i,t}, u_{j,t}\} = \{u, v\}\}, \quad \text{the two cities in rival unions } u \neq v. \quad (4)$$

Table 2: Estimating specifications. Each row is a separate regression with pair and grain-by-year fixed effects and dyad-robust standard errors. “Never co-members” excludes pairs that are within-union partners at any point in the window; “never at a frontier” excludes pairs that are ever in two different unions.

| Specification         | Indicator(s)                                                                    | Additional sample restriction                           |
|-----------------------|---------------------------------------------------------------------------------|---------------------------------------------------------|
| Pooled, same union    | SameUnion $_{ij,t}$                                                             | none                                                    |
| Pooled, union border  | OneInUnion $_{ij,t}$                                                            | never co-members                                        |
| CGCU within-union     | Within $^{CGCU}$                                                                | never co-members of SGCU or PHC;<br>never at a frontier |
| SGCU within-union     | Within $^{SGCU}$                                                                | never co-members of CGCU or PHC;<br>never at a frontier |
| Clean borders         | Border $^{CGCU}$ , Border $^{SGCU}$ ,<br>Border $^{PHC}$                        | never co-members                                        |
| Cross-union frontiers | Frontier $^{CGCU-PHC}$ ,<br>Frontier $^{CGCU-SGCU}$ ,<br>Frontier $^{PHC-SGCU}$ | never co-members                                        |

The regressions estimated are:

$$G_{ij,g,t} = \beta'_s \mathbf{D}_{ij,t}^s + \alpha_{ij} + \delta_{g,t} + \varepsilon_{ij,g,t}, \quad (i, j) \in \mathcal{S}_s, \quad (5)$$

where  $\mathbf{D}^s$  is the indicator or set of indicators belonging to specification  $s$  and  $\mathcal{S}_s$  is its sample. Table 2 lists the six specifications. All of them additionally impose the common restrictions of Section 4.1: cross-state pairs only, within 200 km, 1818–1833, spelt excluded. In every case the omitted category is the pair whose status is unaffiliated in the year concerned, and before 1828 every pair in every sample is in that category.

Separate estimation is necessary because of the reference group. A single regression carrying all nine indicators compares each treated pair type against one residual category mixing genuinely unaffiliated pairs with pairs treated on some other margin; restricting each sample to pairs never touched by another union relationship makes the comparison interpretable, at the cost of power. The last two rows share a sample: what separates the clean-border contrast from the frontier contrast is which indicator is switched on, not a further restriction.

The PHC within-union effect is not estimated at all. Data availability limits the cities in the PHC to Prussian and Hesse-Kassel cities only; no cities from Hesse-Darmstadt are available, and Hesse-Kassel itself is only part of the PHC from 1832 onwards, after the defection, so any estimate would be that single episode in disguise. Entering the unions as separate indicators rather than one customs-union dummy is in any case necessary when the arrangements are contemporaneous and competing (Wolf and Ritschl, 2011).

Every specification carries pair fixed effects  $\alpha_{ij}$ , which absorb all time-invariant pair characteristics including bilateral distance, and grain-by-year effects  $\delta_{g,t}$ , which absorb

the common price level and grain-specific shocks. Identification is within-pair over time, relative to pairs in the same sample whose customs status did not change. Standard errors are dyad-robust (Aronow et al., 2015; Cameron and Miller, 2023), allowing arbitrary correlation across all pairs sharing a city; where the estimator’s finite-sample requirements fail, I report that rather than an unstable number (Section 7.2).

For timing, each specification is re-estimated on its own sample with the treatment indicator replaced by its interactions with year dummies, omitting 1827, the last pre-union year. The indicator is taken in its “ever” form, so that a pair is assigned to the treated group for the whole window on the basis of the status it eventually holds, and the year dummies rather than the indicator carry the timing:

$$G_{ij,g,t} = \sum_{\tau \neq 1827} \gamma_{\tau} (D_{ij} \times \mathbf{1}\{t = \tau\}) + \alpha_{ij} + \delta_{g,t} + \varepsilon_{ij,g,t}, \quad (6)$$

so that  $\gamma_{\tau}$  traces the pair-type’s gap in year  $\tau$  relative to 1827. Coefficients for  $\tau < 1828$  test whether a difference already existed before the treaties.

### 5.3 Spatial statistics

Alongside the dyadic regressions I compute global and local Moran’s I and Geary’s C (Anselin, 1988, 1995) on city price innovations under alternative neighbourhood definitions: geographic (distance bands,  $k$ -nearest neighbours, territorial contiguity) against institutional (shared customs-union membership). The question is which definition better predicts observed co-movement, year by year.

The statistics are computed on price deviations rather than price levels. For each city, grain, and year I take the log price net of that year’s German mean, removing the common macro component (harvest failures, the general movement of the price level) before any spatial statistic is calculated. Without this step the measures would largely record the fact that famine years move every city together, which is not market structure. Four neighbourhood definitions are used. Three are geographic: a binary distance band joining all city pairs within 150 km; a  $k$ -nearest-neighbour matrix with  $k = 4$ , which guarantees no city is isolated; and territorial contiguity, joining cities in the same state whose GEO polygons share a boundary. The fourth is institutional: cities are neighbours if they are co-members under the 1828 union assignment, held constant across the whole period so that the weighting scheme itself does not change when the treaties are signed. Every matrix is row-standardised, and in each year it is first subset to the cities reporting a price for that grain, so that coverage differences across years do not enter through the weights; year–grain cells with fewer than five reporting cities are dropped. Inference is under the normality assumption rather than randomisation.

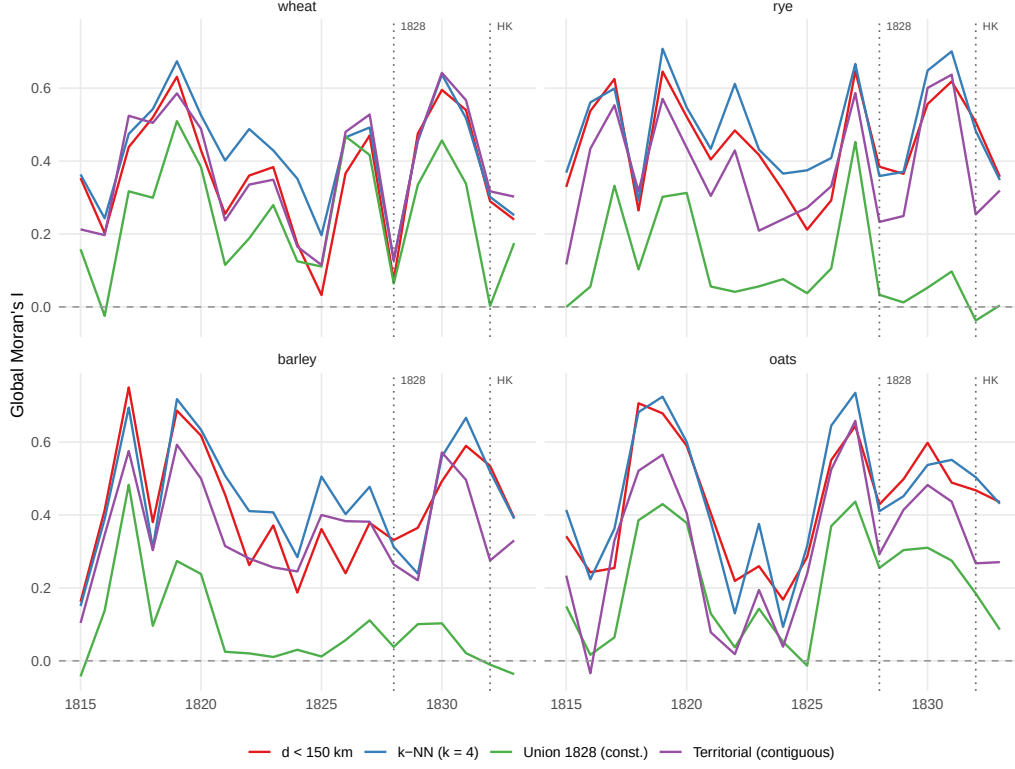


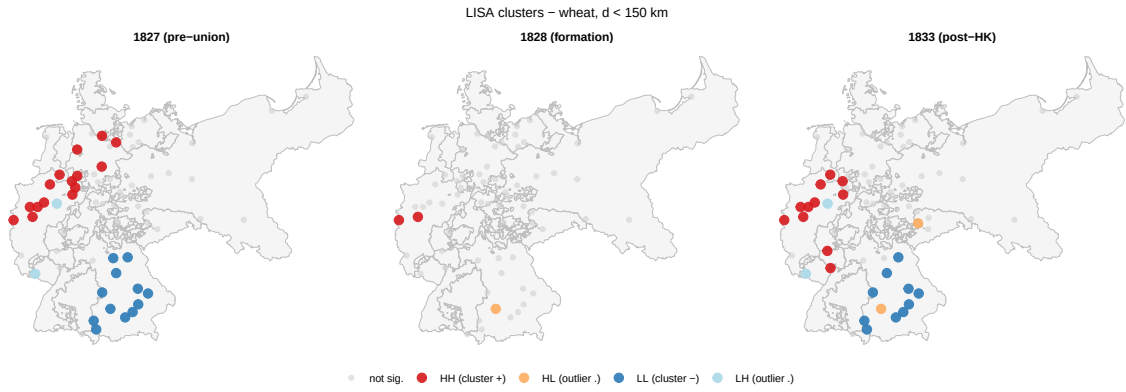
Figure 3: Global Moran's I by year, geographic versus union-membership neighbourhood definitions. Computed on log grain prices net of the annual German mean. Weighting schemes: 150 km distance band, four nearest neighbours, territorial contiguity, and 1828 union co-membership held constant across the period. All matrices are row-standardised and subset each year to reporting cities.

## 6 Spatial Evidence

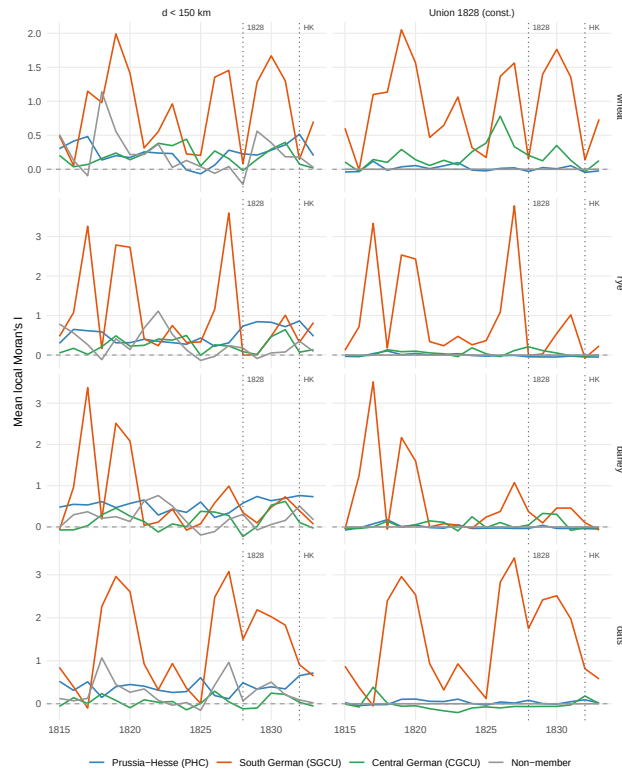
### 6.1 Spatial autocorrelation

Figure 3 reports global Moran's I year by year under geographic and institutional neighbourhood definitions. Averaged over grains and years, the statistic is 0.42 under a 150 km distance band, 0.45 under four nearest neighbours, and 0.36 under territorial contiguity, against 0.16 when fellow union members are treated as neighbours. The institutional definition lies below the average of the three geographic definitions in every year of the period, before 1828 as well as after; the ordering never reverses. All four specifications decline between 1827 and 1828: the institutional one from 0.35 to 0.10, but the geographic ones by a comparable absolute margin, from 0.54 to 0.31 under the distance band and from 0.54 to 0.23 under contiguity. The 1828 decline is therefore a property of that year across all weighting schemes, not of the institutional scheme in particular.

The local statistics show the same pattern. The LISA maps (Figure 4) show clustering organised along geographic lines, with no visible reorganization at union boundaries; CGCU territory exhibits the lowest local clustering of any union throughout the period.



(a) LISA clusters



(b) Local Moran's I over time

Figure 4: Local spatial structure. Panel (a) maps LISA cluster types in 1827, 1828 and 1831 panel (b) plots mean local Moran's I over time by 1828 union status (PHC, SGCU, CGCU or none). Both are computed on log prices net of the annual German mean.

Markets in the western provinces of Prussia appear to be spatially dependent both before and after the formation of the customs unions; interestingly, the parts of the cluster that are inside future members of the CGCU are no longer part of the cluster after union formation. The south also appears to be a cluster around Bavaria, which is present both prior and post union formation.

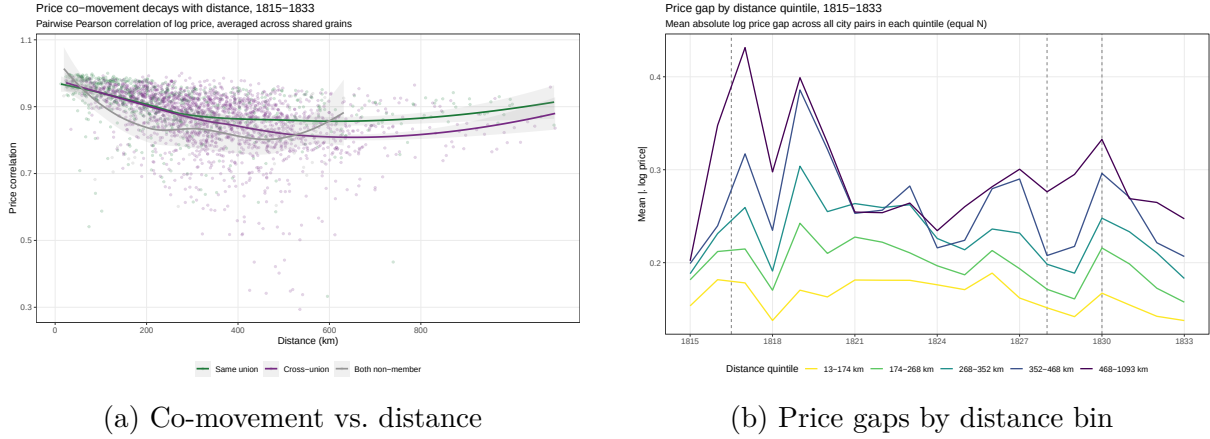


Figure 5: Distance decay of integration measures. Panel (a) plots price-gap co-movement against bilateral distance; panel (b) plots the mean absolute log price gap by distance bin. Sample: cross-state city pairs, 1818–1833, spelt excluded.

## 6.2 Distance

The regressions include city-pair fixed effects and thus do not allow the inclusion of distance within the analysis. It is nonetheless crucial to understand the role of this variable, especially given the costs of transportation that existed in the studied period (pre-railway). I look at two measures that incorporate distance: price series correlation over time between cities, ordered by the distance between those cities, and cross-sectional means of price gaps by distance buckets.

Price-gap correlations decay smoothly with distance (Figure 5), with no visible discontinuity at union borders; pairs at a given distance look alike whether or not a customs boundary runs between them. Price gaps are also strongly related to distance, with the closer pairs exhibiting lower gaps across all years in the sample.

Two cautions belong with this figure. Distance proxies agroclimatic similarity as well as trade friction, so the decay is not evidence about the trade channel specifically; Section 8.4 returns to this. And straight-line distance mismeasures effective distance where water carriage was available, which in this sample means the Rhine, Main, Elbe, and Oder corridors. Nonetheless, these two findings indicate the role that distance played, and combined with the prior results on spatial indicators, show that geography played a strong role in determining markets during this period.

## 7 Regression Results

All tables use the  $\leq 200$  km sample, pair and grain-by-year fixed effects, and dyad-robust standard errors; Table 10 varies the distance threshold for every coefficient. Significance codes in the regression tables are \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ ; the distance sweeps in Tables 10 and 9 use \*\*\*  $p < 0.001$ , \*\*  $p < 0.01$ , \*  $p < 0.05$ , and  $\cdot$   $p < 0.1$ . Significance

levels stated in the text are computed from the estimates directly.

## 7.1 Pooled and within-union effects

Throughout this section, coefficients are reported with their significance level; magnitudes are given only where the estimate is distinguishable from zero. Point estimates and standard errors for every coefficient, significant or not, are in the tables.

Table 4 pools across unions. Sharing *any* union leaves no detectable difference in price gaps: the pooled same-union coefficient is not distinguishable from zero. Table 5 separates the unions. The CGCU within-union coefficient is likewise not distinguishable from zero at the default radius. The SGCU coefficient is positive and significant at the one per cent level, at 0.0426 ( $\approx +4.4\%$ ): SGCU membership is associated with *wider* internal gaps, and the coefficient is significant at every distance threshold (Table 10). It rests on four pairs and five cities, with a single city common to all four pairs (Table 1); Section 8.3 returns to it. The PHC within-union effect is not estimated, for the data reasons set out in Section 5.2: the only cross-state PHC pairs are the Hesse-Kassel pairs after the defection. No union shows a negative within-union coefficient.

Two checks qualify these estimates further. Grain-level estimates (Table 3) show the CGCU null is not an artifact of pooling: barley, oats and wheat show nothing, and only rye is positive and significant. And the SGCU widening is confined to the cross-state pairs: when same-state pairs are retained in the sample (Table 9), so that Bavarian and Württemberg internal pairs also count as SGCU pairs, the coefficient falls to 0.0121 and is no longer distinguishable from zero. This is consistent with the narrow scope already indicated by the pair count.

## 7.2 Clean border effects

The first column of Table 7 compares pairs straddling a union’s border (one member city, one unaffiliated city) with wholly unaffiliated pairs. The point estimates are negative for all three unions, but only one is distinguishable from zero. The CGCU border coefficient is significant at the one per cent level, at  $-0.0257$  ( $\approx -2.5\%$ ). The PHC border coefficient is not distinguishable from zero. The SGCU border coefficient is unreportable as the dyad-robust variance estimator requires more distinct cities than the eight feeding this coefficient, and returns a negative estimated variance; rather than substitute a different estimator for one cell, I report the event-study version below, which is well behaved.<sup>2</sup> A negative border coefficient means border pairs have narrower gaps than unaffiliated pairs at comparable distance. Only the CGCU border effect is estimated with enough precision

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<sup>2</sup>Mechanically, the estimator subtracts a term proportional to the number of distinct cities minus two from a per-city sum of score contributions; with eight cities and eight pairs the subtraction dominates. This is a finite-sample breakdown, not a data problem (Aronow et al., 2015).

Table 3: CGCU within-union effects by grain. Columns: (1) barley, (2) oats, (3) rye, (4) wheat.

| Dependent Variable:                               | Abs. log price gap |                     |                       |                    |
|---------------------------------------------------|--------------------|---------------------|-----------------------|--------------------|
| Model:                                            | (1)                | (2)                 | (3)                   | (4)                |
| <i>Variables</i>                                  |                    |                     |                       |                    |
| Both CGCU                                         | 0.0114<br>(0.0437) | -0.0103<br>(0.0280) | 0.0489***<br>(0.0154) | 0.0067<br>(0.0194) |
| <i>Fixed-effects</i>                              |                    |                     |                       |                    |
| Pair                                              | Yes                | Yes                 | Yes                   | Yes                |
| Year                                              | Yes                | Yes                 | Yes                   | Yes                |
| <i>Fit statistics</i>                             |                    |                     |                       |                    |
| Observations                                      | 1,979              | 1,806               | 2,742                 | 2,319              |
| R <sup>2</sup>                                    | 0.41370            | 0.51570             | 0.37435               | 0.49573            |
| <i>Dyad-robust standard-errors in parentheses</i> |                    |                     |                       |                    |
| <i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i> |                    |                     |                       |                    |

Table 4: Pooled union effects

| Dependent Variable:                               | Abs. log price gap |                       |
|---------------------------------------------------|--------------------|-----------------------|
|                                                   | Pooled: same union | Pooled: union border  |
| Model:                                            | (1)                | (2)                   |
| <i>Variables</i>                                  |                    |                       |
| Same union (any)                                  | 0.0072<br>(0.0110) |                       |
| Union border (any)                                |                    | -0.0212**<br>(0.0083) |
| <i>Fixed-effects</i>                              |                    |                       |
| Pair                                              | Yes                | Yes                   |
| Year-Grain                                        | Yes                | Yes                   |
| <i>Fit statistics</i>                             |                    |                       |
| Observations                                      | 18,772             | 16,263                |
| R <sup>2</sup>                                    | 0.29879            | 0.29006               |
| <i>Dyad-robust standard-errors in parentheses</i> |                    |                       |
| <i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i> |                    |                       |

to say so. By grain (Table 6), these estimates are noisy and mixed in sign: the negative border coefficients concentrate in barley and oats, while rye runs positive for the CGCU and PHC borders. The grain-level cells are small and I do not put weight on them.

Table 5: Within-union effects, by union

| Dependent Variable:                               | Abs. log price gap |                       |
|---------------------------------------------------|--------------------|-----------------------|
|                                                   | CGCU               | SGCU                  |
| Model:                                            | (1)                | (2)                   |
| <i>Variables</i>                                  |                    |                       |
| Both CGCU                                         | 0.0223<br>(0.0144) |                       |
| Both SGCU                                         |                    | 0.0426***<br>(0.0106) |
| <i>Fixed-effects</i>                              |                    |                       |
| Pair                                              | Yes                | Yes                   |
| Year-Grain                                        | Yes                | Yes                   |
| <i>Fit statistics</i>                             |                    |                       |
| Observations                                      | 8,846              | 7,576                 |
| R <sup>2</sup>                                    | 0.30638            | 0.30958               |
| <i>Dyad-robust standard-errors in parentheses</i> |                    |                       |
| <i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i> |                    |                       |

### 7.3 The competitive frontier

The second column of Table 7 reports the frontiers between pairs of rival blocs. The CGCU–PHC coefficient is positive and significant at the one per cent level, at 0.0273 ( $\approx +2.8\%$ ): pairs straddling the frontier between the CGCU and the Prussian system have wider price gaps than unaffiliated pairs at comparable distance. Table 10 varies the radius. CGCU–PHC is positive and significant at every radius reported: at the ten per cent level at 150 km, at one per cent at 200 km, at five per cent at 300 km, and at one per cent in the unrestricted sample, where it is largest at 0.0432 ( $\approx +4.4\%$ ). CGCU–SGCU and PHC–SGCU are significant only within the 150–200 km band, where they reach  $-0.0247$  ( $\approx -2.4\%$ ) and  $0.0294$  ( $\approx +3.0\%$ ) respectively at 200 km; at 300 km and in the full sample neither is distinguishable from zero.

The frontier estimate passes two further checks. By grain (Table 8), CGCU–PHC is positive in every grain and significant in three; wheat is the exception. And it is insensitive to the same-state exclusion: retaining same-state pairs (Table 9) gives 0.0261 at 200 km, significant at the one per cent level, with the same pattern across the distance sweep. Of all the coefficients in this paper, the CGCU–PHC frontier is the one that holds up under nearly every variation tried, with the wheat cell the single exception.

Table 6: Clean border effects by grain. Columns: (1) barley, (2) oats, (3) rye, (4) wheat.

| Dependent Variable:<br>Model:                     | (1)                    | Abs. log price gap     |                      |                    |
|---------------------------------------------------|------------------------|------------------------|----------------------|--------------------|
|                                                   |                        | (2)                    | (3)                  | (4)                |
| <i>Variables</i>                                  |                        |                        |                      |                    |
| CGCU border                                       | -0.0339**<br>(0.0162)  | -0.0401***<br>(0.0094) | 0.0508*<br>(0.0301)  | 0.0124<br>(0.0187) |
| SGCU border                                       | -0.1435***<br>(0.0164) | 0.0380<br>(unstable)   | -0.0264<br>(0.0351)  | 0.0079<br>(0.0315) |
| PHC border                                        | -0.0347**<br>(0.0155)  | -0.0232*<br>(0.0120)   | 0.0773**<br>(0.0367) | 0.0165<br>(0.0172) |
| <i>Fixed-effects</i>                              |                        |                        |                      |                    |
| Pair                                              | Yes                    | Yes                    | Yes                  | Yes                |
| Year                                              | Yes                    | Yes                    | Yes                  | Yes                |
| <i>Fit statistics</i>                             |                        |                        |                      |                    |
| Observations                                      | 1,781                  | 1,641                  | 2,099                | 1,914              |
| R <sup>2</sup>                                    | 0.42209                | 0.52004                | 0.40260              | 0.49250            |
| <i>Dyad-robust standard-errors in parentheses</i> |                        |                        |                      |                    |
| <i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i> |                        |                        |                      |                    |

## 7.4 Timing

Do any of these differences *begin* in 1828? Figure 6 reports the event-study versions of the headline coefficients (equation (6)), with 1827 as the reference year; full coefficient tables are available from the author. No series breaks at 1828.

Two features are worth drawing out. The CGCU–PHC frontier cannot be dated: no year coefficient reaches the five per cent level, so the level difference of Section 7.3 is real while its timing is not recoverable at this radius. And where movement does occur it clusters on 1831–1833 rather than on the treaties. The SGCU within-union widening appears only in 1832 and 1833, and the CGCU–SGCU frontier, having returned to the reference in 1829–1830, separates again from 1831, both coinciding with Hesse-Kassel’s defection and the negotiations that produced the Zollverein. The CGCU border effect runs the other way: it differs significantly from the reference in several pre-treaty years and in none after 1829, so the convergence it records predates the coalition.

## 8 Discussion

Three things follow from the results. First, geography rather than customs membership organises price co-movement across the whole period: the geographic weighting schemes dominate the institutional one in every year, the ordering never reverses, and price gaps

Table 7: Border effects: clean borders and cross-union frontiers

| Dependent Variable:<br>Model:                     | Abs. log price gap               |                             |
|---------------------------------------------------|----------------------------------|-----------------------------|
|                                                   | Border (vs. unaffiliated)<br>(1) | Cross-union frontier<br>(2) |
| <i>Variables</i>                                  |                                  |                             |
| CGCU border                                       | -0.0257***<br>(0.0069)           |                             |
| SGCU border                                       | -0.0535<br>(unstable)            |                             |
| PHC border                                        | -0.0141<br>(0.0116)              |                             |
| CGCU–PHC                                          |                                  | 0.0273***<br>(0.0093)       |
| CGCU–SGCU                                         |                                  | -0.0247**<br>(0.0121)       |
| PHC–SGCU                                          |                                  | 0.0294**<br>(0.0127)        |
| <i>Fixed-effects</i>                              |                                  |                             |
| Pair                                              | Yes                              | Yes                         |
| Year-Grain                                        | Yes                              | Yes                         |
| <i>Fit statistics</i>                             |                                  |                             |
| Observations                                      | 16,263                           | 16,263                      |
| R <sup>2</sup>                                    | 0.29045                          | 0.29144                     |
| <i>Dyad-robust standard-errors in parentheses</i> |                                  |                             |
| <i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i> |                                  |                             |

decay with distance in a way that shows no break at union boundaries. Second, none of the three arrangements integrated the market inside its own territory: there is no negative within-union coefficient anywhere in the paper. Third, the one robust difference connected to policy is not inside a union but between two of them, at the CGCU–PHC frontier. The rest of this section reads that frontier against the treaties, addresses why a tariff channel is unlikely to be what produced it, and sets out what the SGCU result and the design as a whole cannot settle.

## 8.1 The frontier read through the treaties

The institutional content of the CGCU, set out in Section 2.3, is unusual among customs arrangements in that almost none of it is about tariffs among its members. Article XIV's duty-free list was real but reached only goods carried between member states without crossing foreign territory, and for grain it operated solely through weekly-market sales under certificates of origin. Article XIV.2 set a ceiling of half a good-groschen per hundredweight

Table 8: Cross-union frontier effects by grain. Columns: (1) barley, (2) oats, (3) rye, (4) wheat.

| Dependent Variable:<br>Model:                     | (1)                 | (2)                   | (3)                   | (4)                  |
|---------------------------------------------------|---------------------|-----------------------|-----------------------|----------------------|
| <i>Variables</i>                                  |                     |                       |                       |                      |
| CGCU–PHC                                          | 0.0233*<br>(0.0141) | 0.0306**<br>(0.0121)  | 0.0471***<br>(0.0135) | 0.0028<br>(0.0175)   |
| CGCU–SGCU                                         | 0.0209<br>(0.0157)  | -0.0498**<br>(0.0231) | -0.0076<br>(0.0169)   | -0.0452*<br>(0.0241) |
| PHC–SGCU                                          | 0.0279<br>(0.0211)  | -0.0063<br>(0.0144)   | 0.0528**<br>(0.0245)  | 0.0314<br>(0.0238)   |
| <i>Fixed-effects</i>                              |                     |                       |                       |                      |
| Pair                                              | Yes                 | Yes                   | Yes                   | Yes                  |
| Year                                              | Yes                 | Yes                   | Yes                   | Yes                  |
| <i>Fit statistics</i>                             |                     |                       |                       |                      |
| Observations                                      | 3,817               | 3,296                 | 4,789                 | 4,361                |
| R <sup>2</sup>                                    | 0.37694             | 0.49781               | 0.34401               | 0.54257              |
| <i>Dyad-robust standard-errors in parentheses</i> |                     |                       |                       |                      |
| <i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i> |                     |                       |                       |                      |

Table 9: Distance-threshold robustness with same-state pairs retained. Same specifications as Table 10, with pairs of cities in the same state kept in the estimation samples.

| Coefficient        | 100km               | 150km               | 200km               | 300km             | Full sample        |
|--------------------|---------------------|---------------------|---------------------|-------------------|--------------------|
| Same union (any)   | 0.0086 (0.0061)     | 0.0035 (0.0063)     | 0.0062 (0.0057)     | 0.0195** (0.0061) | 0.0423*** (0.0069) |
| Union border (any) | -0.0260* (0.0103)   | -0.0161 (0.0113)    | -0.0199* (0.0082)   | -0.0194. (0.0112) | -0.0154. (0.0092)  |
| Both CGCU          | 0.0247* (0.0107)    | 0.0167 (0.0122)     | 0.0179 (0.0117)     | 0.0207 (0.0150)   | 0.0219 (0.0169)    |
| Both SGCU          | 0.0232. (0.0123)    | 0.0104 (0.0128)     | 0.0121 (0.0105)     | 0.0225 (0.0145)   | 0.0270* (0.0136)   |
| CGCU border        | -0.0200 (0.0127)    | -0.0191. (0.0101)   | -0.0245*** (0.0065) | -0.0205* (0.0088) | -0.0261* (0.0125)  |
| SGCU border        | —                   | -0.0681*** (0.0019) | -0.0450*** (0.0046) | -0.0174. (0.0095) | -0.0225** (0.0085) |
| PHC border         | -0.0239*** (0.0042) | -0.0022 (0.0175)    | -0.0130 (0.0116)    | -0.0188 (0.0147)  | -0.0075 (0.0121)   |
| CGCU–PHC           | —                   | 0.0216. (0.0112)    | 0.0261** (0.0091)   | 0.0281* (0.0116)  | 0.0424*** (0.0104) |
| CGCU–SGCU          | —                   | -0.0560*** (0.0143) | -0.0257* (0.0117)   | -0.0193 (0.0200)  | -0.0236. (0.0138)  |
| PHC–SGCU           | —                   | 0.0307. (0.0169)    | 0.0283* (0.0124)    | 0.0150 (0.0144)   | 0.0038 (0.0118)    |

on transit duties where such duties already existed, which is a cap on pre-existing charges rather than their removal. What the treaty bound unconditionally was external: every member undertook not to accede to either rival union for the treaty’s duration. The coalition’s operative commitment, in other words, was to hold its members outside the Prussian tariff of 1818, a schedule that was already a decade old when the coalition formed, and which the coalition neither raised nor lowered.

Before interpreting the frontier gap in these terms, it is worth stating what would have contradicted this reading. If blocked accession had economic consequences at the margin where it bound, we should find a persistent positive gap specifically at the CGCU–PHC frontier, and we should not find the same thing at frontiers where nothing comparable

Table 10: Distance-threshold robustness, all coefficients

| Coefficient        | 150km               | 200km               | 300km              | Full sample        |
|--------------------|---------------------|---------------------|--------------------|--------------------|
| Same union (any)   | 0.0029 (0.0094)     | 0.0072 (0.0110)     | 0.0238** (0.0109)  | 0.0276*** (0.0097) |
| Union border (any) | -0.0181 (0.0116)    | -0.0212** (0.0083)  | -0.0191* (0.0112)  | -0.0150 (0.0095)   |
| Both CGCU          | 0.0157 (0.0175)     | 0.0223 (0.0144)     | 0.0362** (0.0165)  | 0.0328* (0.0179)   |
| Both SGCU          | 0.0304*** (0.0116)  | 0.0426*** (0.0106)  | 0.0451*** (0.0117) | 0.0465*** (0.0095) |
| CGCU border        | -0.0212** (0.0106)  | -0.0257*** (0.0069) | -0.0205** (0.0088) | -0.0259** (0.0126) |
| SGCU border        | —                   | —                   | -0.0121* (0.0073)  | -0.0221** (0.0089) |
| PHC border         | -0.0039 (0.0174)    | -0.0141 (0.0116)    | -0.0186 (0.0148)   | -0.0073 (0.0122)   |
| CGCU–PHC           | 0.0227* (0.0124)    | 0.0273*** (0.0093)  | 0.0293** (0.0120)  | 0.0432*** (0.0105) |
| CGCU–SGCU          | -0.0553*** (0.0153) | -0.0247** (0.0122)  | -0.0182 (0.0202)   | -0.0229 (0.0140)   |
| PHC–SGCU           | 0.0321* (0.0180)    | 0.0294** (0.0127)   | 0.0162 (0.0138)    | 0.0045 (0.0120)    |

Dyad-robust standard errors in parentheses. Signif. codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1.

was at stake. Three patterns would have refuted that. A null or negative CGCU–PHC coefficient would have said the blocking commitment had no measurable price effect. A CGCU–PHC coefficient no more robust than the other two cross-union coefficients would have said that frontiers between rival blocs look alike in general, and that nothing specific to this rivalry was operating. And a comparably robust gap at the CGCU–SGCU frontier, where the two blocs were rivals but no accession was being actively blocked in the same way, would have pointed to a generic feature of bloc boundaries rather than to the no-join clause.

None of the three is observed. The CGCU–PHC coefficient is positive and significant at every radius reported, and it is the only cross-union estimate that survives the full distance sweep; the CGCU–SGCU and PHC–SGCU coefficients are significant within a narrow band and indistinguishable from zero on either side of it. The comparison across the three frontiers, rather than the before-and-after comparison at 1828, is what supports the reading. This matters because the absence of a break at 1828 cannot by itself discriminate: the geographic account of Section 6 predicts exactly the same absence, and a result consistent with two rival readings favours neither.

A blocked accession, on its own, would explain an absence of narrowing rather than an excess gap. Two features of this particular frontier bear on why the gap is positive rather than merely unchanged. The first is that accession was live here and nowhere else. Hesse-Darmstadt acceded to the Prussian system in February 1828; Hesse-Kassel, similar in size, situation and commercial orientation, was held outside it by the no-join clause until August 1831, when it left the coalition at the first opportunity. The barrier removal that did not happen at this frontier is one that was actively under negotiation. The second is that the coalition’s territory sat between the two Prussian territorial blocks, and Hesse-Kassel levied transit charges on goods moving between Prussia’s Rhenish provinces and its eastern lands (Hahn, 1984). The frontier was therefore not only a line across which accession was blocked but a margin on which one side drew revenue from the other’s internal commerce.

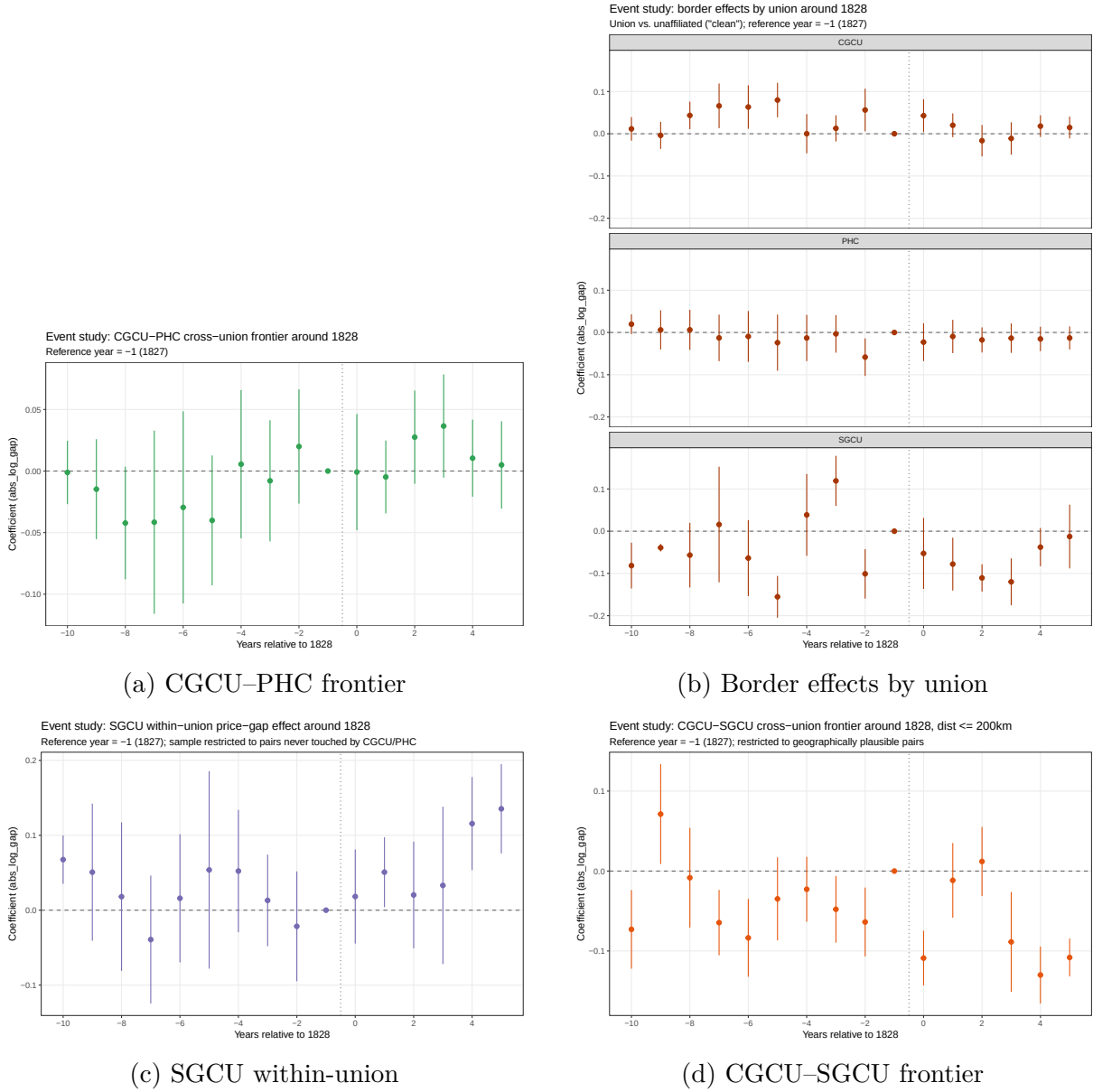


Figure 6: Event studies, reference year 1827. Each panel plots the coefficients  $\gamma_\tau$  from equation (6): the pair type's absolute log price gap in year  $\tau$  relative to 1827, with pair and grain-by-year fixed effects and dyad-robust standard errors. Sample: cross-state pairs within 200 km, 1818–1833.

The contemporary record is consistent with both sides treating this margin as contested. Motz's stated conditions for the negotiations with the south included the elimination of the coalition's negative effects on Prussia as a named objective (Ploeckl, 2015, p. 285), and from the autumn of 1828 Prussia financed toll-free bypass roads through the Thuringian states in order to route transit away from Hesse-Kassel (Hahn, 1984). Austrian support for the coalition, by contrast, was declaratory: the material commercial concessions its members hoped for did not arrive.

Two limits on this reading should be stated plainly. The timing of the frontier gap cannot be established: the event study does not resolve when the difference arose (Section 7.4),

so the argument rests on the level of the gap and on the contrast across frontiers, not on any dating. And the design is descriptive throughout. Membership was not assigned; the coalition formed among states whose geography and commercial position were exactly what made them worth blocking. Nothing here identifies the causal effect of the no-join clause, and the association is reported as an association.

## 8.2 Tariff magnitudes and the channel question

A natural objection is that customs unions work through tariffs, so a price study of grain should be looking for tariff changes, and that if none is found, nothing has been shown. For this episode the premise fails, because the grain-specific tariff content of all three arrangements was small.

The Prussian–Hessian system extended Prussia’s own 1818 schedule to a new partner rather than negotiating a new one, and that schedule’s nominal grain rates were low, even as weight-based duties and falling prices raised effective protection on other goods over the following decade (Hahn, 1982; Oechelhäuser, 1851). The CGCU reduced grain duties only in the narrow channel described above and capped, rather than removed, existing transit charges. The SGCU is the only one of the three that dissolved internal customs posts outright. Waterborne traffic, which carried much of the bulk grain trade, lay outside all three treaties: Rhine and Elbe shipping were governed by separate commissions established at the Congress of Vienna (Ploeckl, 2021).

If the tariff margin was largely untouched for grain, then a tariff channel is not what these unions could plausibly have transmitted into grain prices, and the absence of within-union convergence is what the treaty content leads one to expect rather than a puzzle. What the arrangements did carry was routing, transit and commitment. The one margin where the price evidence is not null is the frontier where these were contested.

## 8.3 The SGCU anomaly

The South German union does not fit either account, and I do not resolve it. It was contiguous, negotiated between approximate equals, and the only one of the three to abolish internal customs posts. On the geographic reading it should have shown the clearest internal integration; on the institutional reading it should have shown the clearest treaty effect. Instead its within-union coefficient is positive and significant at every distance threshold: member cities’ price gaps are wider, not narrower, than their own pre-union baseline.

The scope of that result should be stated before anything is built on it. It rests on four pairs and five cities, with a single city common to all four pairs. The coefficient is statistically robust and substantively narrow: it describes one city’s relationships with its neighbours, and the sample cannot establish whether it describes more than that.

The event study adds one fact that a purely mechanical explanation would not predict. The widening does not appear in 1828. It appears in 1832 and 1833, after Hesse-Kassel’s accession to the Prussian system and during the negotiations that produced the Zollverein. One interpretive possibility, which I offer rather than assert, is that this reflects the post-1831 reconfiguration of northward routes: with Hesse-Kassel inside the Prussian system, southern traffic moving north faced Prussian transit charges on routes that had not carried them before, a pressure that has been identified as one reason Bavaria and Württemberg entered the 1833 negotiations (Ploeckl, 2015, p. 292). The timing matches. Four pairs cannot carry the claim, and I leave it open.

## 8.4 Limitations

Four limits bound what can be concluded. Distance proxies agroclimatic similarity as well as trade friction, as nearby cities share weather, and grain prices respond to it (Albers and Pfister, 2021). Straight-line distance also mismeasures effective distance where water carriage was available, and navigation on the Elbe and the Rhine was reformed inside the study window, in 1821 and 1831, so transport costs on those corridors were changing by policy. The data are annual and cover one commodity class: within-year arbitrage is invisible, and the unions’ effects on manufactures, where tariff schedules differed far more, are untested. Inference on dyadic data converges in the number of cities, and 60 to 65 is modest (Cameron and Miller, 2023); the two SGCUs, resting on four and eight pairs, are the clearest instance, one of them admitting no valid standard error at all.

Above all, membership was not assigned. The coalitions formed among states whose geography, trade routes and fiscal positions were precisely what made membership attractive or defensive, and those characteristics enter price gaps directly (Keller and Shiue, 2014; Wolf and Ritschl, 2011). No design available here separates the arrangements from the history that produced them, which is why the paper reports associations and reads them against the treaty record.

## 8.5 Future work

Four extensions follow directly. The Hesse-Kassel defection of 1831 moves the CGCU–PHC frontier itself, which makes the post-1832 period a within-frontier test that the present specification pools away; either result would be informative. The Hesse-Darmstadt and Hesse-Kassel accessions offer the sharpest comparison the coalition-externality account implies (two similar states, one admitted in 1828 and one blocked until 1831), and the border pairs needed for it are already in the sample. Beyond this episode, the determinants of who joined which coalition remain contested between fiscal (Dumke, 1984) and geographic or strategic (Ploeckl, 2015) accounts, and the price data assembled here bear on that question indirectly. Lastly, the welfare content of a frontier gap is not established by

the gap itself; linking these price differences to urban population or output over a longer window would give the magnitudes economic meaning.

## 9 Conclusion

In 1828 three customs arrangements formed among the German states within a single year and in competition with one another. The episode is the earliest in which several trade blocs were assembled simultaneously on the same territory, each partly in response to the others, and it ended six years later in the Zollverein. The quantitative literature on German market integration has treated the years before 1834 as a single undifferentiated baseline; this paper has instead measured the three arrangements separately, in grain prices, over 1818–1833.

The results are largely negative where the institutional literature might expect them to be positive, and positive in one place it has not looked. None of the three unions is associated with narrower price gaps among its own members. Geographic proximity predicts which cities' prices move together considerably better than shared membership does, in every year of the period and under every weighting scheme tried. Against that background, one boundary does leave a residual difference that geography does not explain: the frontier between the Central German coalition and the Prussian system, where price gaps are wider by roughly 2.8 per cent, and which is the only cross-union estimate robust to the choice of distance threshold.

Read against the treaties, that pattern is what the coalition's institutional content implies. The Central German agreement removed almost no barriers among its members: its duty-free list for grain operated only through weekly-market sales and only along routes that did not cross foreign territory, a condition its own scattered geography frequently failed. It raised none against Prussia, capping rather than abolishing existing transit charges. What it bound unconditionally was accession: no member could join either rival for the treaty's duration. The frontier where that commitment bound is the frontier where the gap appears, and the frontiers where nothing comparable was at stake show no robust difference. The evidence does not date this to 1828 and cannot; the argument rests on the level of the gap and on the contrast across the three frontiers, not on a break at the treaties.

The wider point concerns what regional agreements do when they form in competition. The trade-policy literature asks whether such agreements are building blocks or stumbling blocks for wider integration, and generally frames the stumbling-block case in terms of barriers erected or trade diverted. The 1828 system points to a different mechanism. The Central German Commercial Union raised no tariffs and diverted no trade by design; its economic content was a commitment by its members not to remove the barriers that already separated them from Prussia. Fragmentation was preserved not by new barriers

but by a contract not to remove existing ones. On the evidence assembled here, that commitment is the only element of the 1828 settlement that shows up in the price data. The least institutionally substantive of the three arrangements is, on this evidence, the one with the clearest measurable consequences.

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## **Data availability**

The grain price series underlying this study derive from Albers and Pfister (2024), Jacks (2005) and Shiue and Keller (2007). The constructed city–year price panel, the territorial (GEO unit) panel, the derived dyadic price-gap file, and the analysis code that reproduces every table and figure in this paper are available from the author on request. A public deposit is planned on publication.

## **Declaration of generative AI and AI-assisted technologies in the manuscript preparation process**

During the preparation of this work the author used Claude (Anthropic) in order to revise sections of the manuscript text, to restructure and edit prose, and to assist with reproducing and cross-checking reported estimates against the analysis output. After using

this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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